SARA COOPER AMUNDARAIN

SOCIAL ROBOTICS RESEARCH ENGINEER

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ABOUT ME

Robotics software and biomedical engineer specializing in the design and deployment of social robots for healthcare. My work spans humanoid platforms for cognitive stimulation and health monitoring, tabletop robots for cross-cultural social mediation, and therapeutic systems for autism intervention. I combine strong technical expertise focused on ROS2 with research experience and project management to bring innovative robotic solutions from concept to real-world deployment.

WORK HISTORY

Research scientist

IIIA-CSIC, Bellaterra, Spain | April 2025 - Present

- Tabletop social robot for autism, featuring a multimodal interaction manager, expressive behaviours, and therapeutic games co-designed with therapists.

Social Robotics Research Engineer

PAL Robotics, Barcelona, Spain | July 2024 - March 2025

- R&D on the tabletop TIAGo Head robot as a social mediator, integrating LLM dialogue, a knowledge base, and ROS2 perception. Product presentation paper at RO-MAN.

Research Scientist

Honda Research Institute Japan, Wako, Japan| April 2023 - February 2024

- Project coordination and user-case development for the Social Mediator project using the tabletop Haru robot in cross-cultural school pilots (Japan–Australia)
- Research and ROS/Python development in socio-emotional interaction design (expressiveness, dialogue), with results presented at RO-MAN, ICRA, HRI.

Robotics Software engineer and ARI Technical Expert

PAL Robotics, Barcelona, Spain | January 2020 - January 2023

- Technical Expert for the ARI social robot, defining product priorities and managing internal, EU, and national projects to ensure high-quality, on-time deliverables; coordination with business (datasheets, client meetings) and marketing
- Technical coordination of healthcare robotics projects (H2020 SPRING, SHAPES, AMIBA), including co-design of ARI social robot use-cases with older adults for cognitive games, reminders, health monitoring, hospital reception, and physical activities in day-care centers.
- ROS2/Python/JavaScript development of dialogue pipelines (RASA, Vosk), Behavior Trees, navigation, and ROS4HRI for human-robot interaction skills of PAL robots.
- Research output on RO-MAN, HRI, ICSR conferences (papers, invited talks)

EDUCATION

Master in Robotics (with Distinction)

Heriot-Watt University, Edimburgh, UK | September 2018 - August 2019

- Acquired skills on Robotics Operating System (ROS), biologically inspired computation, robot kinematics, through group projects and robots like Turtlebot (navigation), Husky with UR5 (manipulation)
- MSc project on object handover with TIAGo robot, collaborating with psychologists of University of Stirling to analyse cognitive load with EEG. User-study with 16 students.

Bachelor in Biomedical Engineering

Mondragon Unibertsitatea, Mondragon, Spain | September 2014 - July 2018

- PBL Project-based learning with group projects involving design of an infant incubator, electrocardiogram system acquisition, hip implant
- Graduate project at Middlesex University (10 months, London), using Pepper robot for cognitive stimulation delivery. User-study with 10 older adults

SKILLS

- Project management
- Critical thinking
- Team work
- Robotics Operating System (ROS1, ROS2)
- Python
- Javascript/HTML

LANGUAGES

EUSKERA, ENGLISH, SPANISH
Native

JAPANESE
Medium

CATALAN
Beginner

RESEARCH PAPERS

The RUSH Checklist: A Standardized Framework for Reporting User Studies in Human–Robot Interaction
Chandra, et al. "The RUSH Checklist: A Standardized Framework for Reporting User Studies in Human–Robot Interaction." Proceedings of the 21st ACM/IEEE International Conference on Human–Robot Interaction (HRI '26). 2026.

EMOROBCARE: A Low-Cost Social Robot for Supporting Children with Autism in Therapeutic Settings
Cooper, S. et al.. "EMOROBCARE: A Low-Cost Social Robot for Supporting Children with Autism in Therapeutic Settings." International Conference on Social Robotics. 2025.

TIAGo Head: an AI Powered Platform for Social Robotics
Cooper, S. et al. "TIAGo Head: an AI Powered Platform for Social Robotics." IEEE International Conference on Robot and Human Interactive Communication (RO-MAN). 2025.

Design of Social Features For Robot-mediated Cross-cultural interaction
Cooper, Sara, et al. "Design of Social Features for Robot-mediated Cross-cultural Interaction." Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction. 2024.

Design of embodied mediator haru for remote cross cultural communication Gomez, Randy, et al. "Design of embodied mediator haru for remote cross cultural communication." 2024 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2024.

Challenges of deploying assistive robots in real-life scenarios: an industrial perspective
Cooper, Sara, Raquel Ros, and Séverin Lemaignan. "Challenges of deploying assistive robots in real-life scenarios: an industrial perspective." 2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO- MAN). IEEE, 2023.

Open-source Natural Language Processing on the PAL Robotics ARI Social Robot
Lemaignan, Séverin, et al. "Open-source Natural Language Processing on the PAL Robotics ARI Social Robot." Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction. 2023.

Towards the deployment of a social robot at an elderly day care facility
Cooper, Sara, and Raquel Ros. "Towards the deployment of a social robot at an elderly day care facility." International Conference on Social Robotics. Cham: Springer Nature Switzerland, 2022.

Towards using behaviour trees for long-term social robot behaviour
Cooper, Sara, and Séverin Lemaignan. "Towards using behaviour trees for long- term social robot behaviour." 2022 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI). IEEE, 2022

Social robotic application to support active and healthy ageing
Cooper, Sara, et al. "Social robotic application to support active and healthy ageing." 2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN). IEEE, 2021.

Assistive technologies for supporting wellbeing of older adults
Dratsiou, Ioanna, et al. "Assistive technologies for supporting wellbeing of older adults." Proceedings of the 14th PErvasive Technologies Related to Assistive Environments Conference. 2021.

ARI: The social assistive robot and companion
Cooper, Sara, et al. "ARI: The social assistive robot and companion." 2020 29th IEEE International conference on robot and human interactive communication (RO-MAN). IEEE, 2020

An EEG investigation on planning human-robot handover tasks
Cooper, Sara, et al. "An EEG investigation on planning human-robot handover tasks." 2020 IEEE International Conference on Human-Machine Systems (ICHMS). IEEE, 2020.